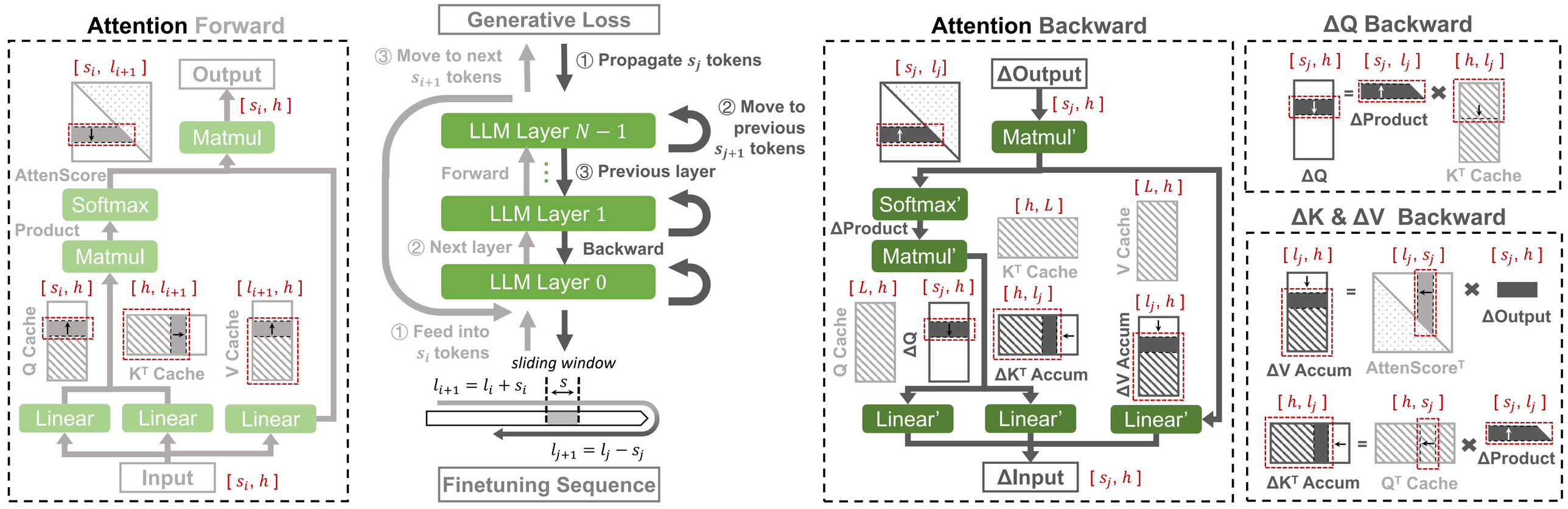

Attention Forward
[s_i, l_{i+1}]
Output
[s_i, h]
Matmul
AttenScore
Softmax
Product
Matmul
[s_i, h]
[h, l_{i+1}]
[l_{i+1}, h]
Q Cache
K^T Cache
V Cache
Linear
Linear
Linear
Input
[s_i, h]
Generative Loss
③ Move to next s_{i+1} tokens
① Propagate s_j tokens
② Move to previous s_{j+1} tokens
LLM Layer $N-1$
Forward
③ Previous layer
LLM Layer 1
② Next layer
Backward
LLM Layer 0
① Feed into s_i tokens
sliding window
$l_{i+1} = l_i + s_i$
s
$l_{j+1} = l_j - s_j$
Finetuning Sequence
Attention Backward
[s_j, l_j]
ΔOutput
[s_j, h]
Matmul'
Softmax'
ΔProduct
Matmul'
[h, L]
K^T Cache
[L, h]
V Cache
[L, h]
[s_j, h]
[h, l_j]
[l_j, h]
Q Cache
ΔQ
ΔK^T Accum
ΔV Accum
Linear'
Linear'
Linear'
ΔInput
[s_j, h]
ΔQ Backward
[s_j, h]
[s_j, l_j]
[h, l_j]
=
ΔProduct
ΔQ
K^T Cache
ΔK & ΔV Backward
[l_j, h]
[l_j, s_j]
[s_j, h]
=
ΔOutput
ΔV Accum
AttenScore^T
[h, l_j]
[h, s_j]
[s_j, l_j]
=
ΔProduct
ΔK^T Accum
Q^T Cache